

Performance Testing

Date	10 July 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Loan Eligibility Prediction
Test Environment	Local System (Flask Server)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid loan details	10	30	Loan prediction displayed successfully

2	Submit invalid input	5	20	Validation error displayed
3	Multiple prediction requests	20	60	System responds without failure
4	Continuous prediction requests	25	90	Stable performance with acceptable response time



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within limit
3	Throughput (Req/sec)	20 Req/sec	22 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	48%	Pass	Normal usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4: Observations & Analysis

The Smart Lender application handled multiple loan prediction requests successfully with low response time and no significant errors. CPU and memory usage remained within acceptable limits, indicating stable system performance.

Step 5: Screenshots / Evidence

Smart Lender - Loan Prediction

Gender (Male=1, Female=0)

Married (Yes=1, No=0)

Dependents

Education (Graduate=1, Not Graduate=0)

Self Employed (Yes=1, No=0)

Applicant Income

Coapplicant Income

Loan Amount

Loan Amount Term